

SHARAN KUMAR U

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SUMMARY

AI Backend engineer with 1.5+ years building production LLM systems, agentic tool-calling backends, multi-provider model orchestration and real-time voice pipelines on FastAPI and PostgreSQL, deployed on GCP. Six production systems shipped across travel, ed-tech and sales-intelligence.

TECHNICAL SKILLS

Languages	Python, TypeScript, SQL, JavaScript
Backend	FastAPI, async SQLAlchemy, Pydantic, PostgreSQL / PostGIS, Redis, Alembic, SSE & WebSocket streaming, pytest
AI & LLM	OpenAI, Anthropic Claude, Gemini, Grok; function calling & tool orchestration, multi-agent pipelines, prompt caching, RAG (ChromaDB), token & cost accounting
Voice / Real-time	LiveKit Agents, Deepgram STT, ElevenLabs TTS, OpenAI TTS/STT, Twilio WhatsApp, Telegram Bot API
Cloud & DevOps	GCP Cloud Run, Cloudflare, Docker & Compose, GitHub Actions, Nginx, uv, Git
Frontend	SvelteKit, Next.js, React, Tailwind CSS, Playwright

WORK EXPERIENCE

TELIC INFO SERVICES PVT LTD

Chennai, IN

AI Developer

May 2025 – Present

- Architected the **AI concierge** for a multi-country travel PWA built by a **5-engineer team** - an 11-tool function-calling agent over FastAPI and PostgreSQL/PostGIS backing **226 REST endpoints**, with SSE streaming surviving client disconnect and 2,100+ backend tests.
- Built an **OpenAI → Gemini provider router** with automatic failover and migrated both providers to the `/v1/responses` API, adding reasoning-effort controls and truncation detection.
- Cut LLM spend through **cross-provider prompt caching** (Anthropic `cache_control`, Gemini explicit caching, OpenAI cached-token discounting) with per-turn token accounting and quota billing persisted to PostgreSQL.
- Delivered **real-time voice** across web, Telegram and WhatsApp using LiveKit Agents with Deepgram STT and ElevenLabs TTS - live transcript streaming, speaker attribution and per-speaker audio lanes.
- Owned **three security-audit remediations** end to end (30+ findings): JWT algorithm enforcement, IDOR ownership checks, CSRF, user-level rate limiting, encrypted secret storage and log sanitisation.
- Deployed to **GCP Cloud Run behind Cloudflare** with Docker Compose, Alembic migrations and GitHub Actions CI; drove Redis write-through caching and N+1 query and index fixes across hot read paths.
- Shipped through **peer code review** across four repositories; reviewed teammates' PRs on shared feature branches.

AI Developer Intern

Dec 2024 – Apr 2025

- Built 5+ AI applications on the **Phidata agentic framework** - a video-to-SOP generator converting tutorial videos into structured Word/PDF procedures, a museum **RAG chatbot** with camera and audio input, and a PRD generator producing architecture and Mermaid diagrams.

SELECTED PROJECTS

Turisqua - AI Travel Concierge | FastAPI, PostGIS, Redis, OpenAI/Gemini, LiveKit, SvelteKit

Built the agent tool layer (place, venue, route and meetup search, price benchmarks, weather, itinerary sequencing), the personalised daily-brief generator, Redis write-through caching, Trust & Safety, and offline road-mode with turn-by-turn navigation.

David - Live-Call Co-Pilot | Python, LiveKit, Deepgram, Next.js, Electron

Real-time coaching agent built on Vexa (Apache-2.0 self-hosted meeting-bot platform), a 3,300-commit codebase - one grounded note per speaker turn. Deterministic intent router, three kind-native agents, a boundary gate blocking pricing and closing advice, and a confidential/speakable brief split comparing digits to prevent numeric leaks. 341 tests; **8 feature PRs merged through peer review**.

Mystic Advisor - Multi-LLM Conversational Backend | FastAPI, asyncpg, Redis, OpenAI/Claude/Gemini

Sole backend owner. Multi-agent pipeline (archetype detection, intent classification, session summarisation), a versioned prompt system, Redis-backed real-time session timers, SSE streaming and token-based billing with race-safe deduction. 526 tests.

Optima - SEO/AEO Audit Platform | FastAPI, Playwright, PostgreSQL, SvelteKit

Full-stack. Crawls up to 50 pages concurrently and runs 38 checks - 33 SEO plus 5 AEO tuned for ChatGPT, Google SGE and Perplexity - with SSRF protection and LLM-generated fixes. Built the SvelteKit dashboard: score gauge, issue heatmap, page-level remediation.

EDUCATION

Vellore Institute of Technology, Chennai - B.Tech, Computer Science and Engineering

2021 – 2025

ACHIEVEMENTS & CERTIFICATIONS

Finalist, Gen-AI Hackathon - top 8 of 98 teams; built a Gen-AI stock advisor on Google and Meta LLMs.

AWS Certified Cloud Practitioner | Machine Learning: Advanced Learning Algorithms (Coursera) | Computer Vision (NPTEL).